

Enhancing Adversarial Robustness of Mobility-Scooter Riding Stability Analysis Model through Body-Informed Design

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Abstract—Motorized mobility scooters provide essential independence for adults with mobility challenges, but consistent in-clinic stability assessment is hard to scale. Skeletal-pose graph convolutional networks (GCNs) classify rider stability accurately on clean inputs, yet small adversarial perturbations of the joint coordinates collapse their predictions. We propose HGCN, a body-informed graph network that fuses a frozen Hard-Constraint Graph Autoencoder (HCGAE) over three anatomical part graphs with a frozen adversarially trained GCN backbone by a small, trainable fusion head. Using real-world mobility-scooter riding pose data with clinically labeled stable/unstable ground truth, our extensive experiments show HGCN has significant improvement of adversarial robustness on rider-stability classification compared to GCN baseline models. At evaluation $\varepsilon = 0.01$, HGCN reaches 0.74 mean adversarial ROC-AUC against four white-box attacks and 0.71 under a leave-one-attack-out protocol, while maintaining clean ROC-AUC at 0.96.

Index Terms—Riding Pose Stability Analysis, Adversarial Robustness, Graph Convolutional Networks.

I. Introduction

Motorized mobility scooters provide essential independence for adults with mobility challenges, yet their safe operation remains an evolving concern that periodic in-clinic assessments cannot fully capture or address [1]. Edge AI applied to a rider’s pose stream could help close this gap by monitoring stability in community environments [2]. Modern deep learning classifiers, such as the graph convolutional network (GCN) [3], have demonstrated high classification accuracy and throughput in some edge-based medical discriminative analytics applications [4]. Despite these promising technical advances, GCN-based pose classifiers remain vulnerable to small adversarial perturbations in their inputs [5]. The stability analysis of mobility scooter riding, which is relevant to safety, requires domain-specific robustness mechanisms rather than defenses directly transferred from image classification.

We propose **HGCN**, a body-informed graph network for robust stability classification under adversarial perturbations to the pose input. HGCN combines HCGAE, a body-structure-informed auto-encoder with an adversarially trained GCN backbone through a lightweight fusion head, allowing the classifier to rely on stable anatomical features when GCN

predictions become brittle. The idea is that body-structure features are more stable than raw pose input under small perturbations. When the GCN backbone becomes sensitive to noisy pose, the fusion head can rely more on the HCGAE features, maintaining clean accuracy while improving robustness.

Our contributions in this paper are threefold. (i) For mobility scooter riding stability classification, we introduce HGCN, a skeletal-pose classification architecture that incorporates anatomical-structure priors as an architectural defense against adversarial perturbations. (ii) Based on mobility scooter riding pose data and clinically labeled ground truth, we show that, unlike a plain GCN whose clean accuracy substantially deteriorates under adversarial training, HGCN preserves near-baseline clean performance while improving robustness. (iii) Experiments that HGCN generalizes to unseen attack configurations under a leave-one-attack-out evaluation protocol.

II. Related Work

Following Kipf and Welling [6], many skeleton-GCN variants are proposed to model anatomical connectivity, temporal motion, and body-part structure, as in GLA-GCN [7]. However, prior work focuses on clean-input accuracy rather than robustness to adversarial perturbations. Adversarial robustness has been studied extensively in the image domain such as the fast gradient-sign attack that motivated the iterative methods that followed [5]. Madry et al. later established multi-step projected gradient descent as a strong attack and demonstrated empirically robust classifiers via mixed clean / adversarial training [8]. HGCN’s design pattern of two frozen pretrained branches with a small trainable fusion head is an instance of parameter-efficient transfer learning, a paradigm surveyed in the visual tuning literature [9]. Smart-health applications prevail on pretrained foundation models adapted with task-specific heads, as outlined in recent surveys of healthcare foundation models [10]. Our work is designed as a body-informed defense for the human-pose based medical applications in the adversarial setting.

III. Methodology

III-A. Data Collection and Preprocessing

We collect data from mobility scooter driving sessions, involving 26 patients with various neurological conditions at

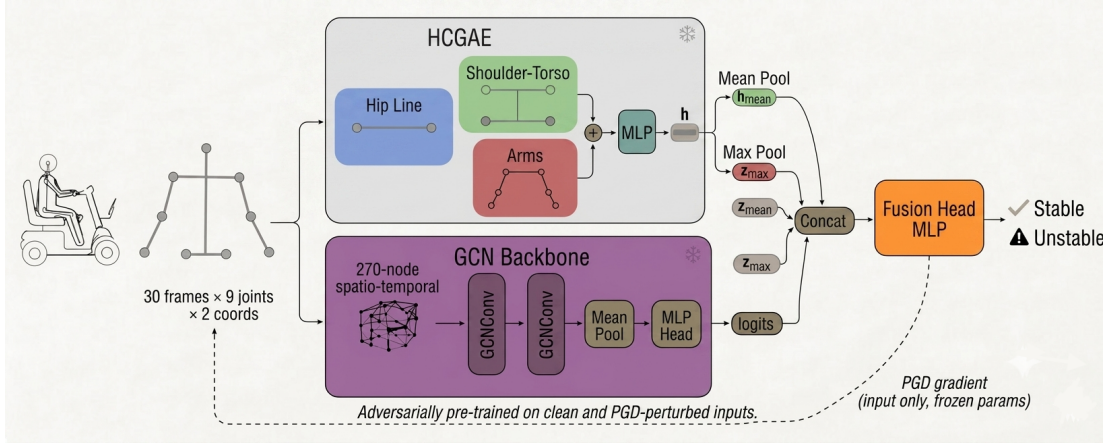


Fig. 1. System overview of HCGCN with Enhanced Adversarial Robustness for Mobility Scooter Riding Stability Classification.

Casa Colina Hospital and Centers for Healthcare in Pomona, CA, under an approved IRB. Front-facing cameras captured participants' upper-body movements at 30 frames per second. In addition, a Raspberry Pi device recorded scooter motion data and maintained a synchronized video stream. In total, we used 502,260 seconds of video recordings for experiments with a 80%/10%/10% split of train/val/test. The front-view videos were processed by a YOLOv7-based pose estimation pipeline [11]. For each frame, the pipeline extracts normalized 2-dimensional coordinates of the upper-body joints, and we use 9 of them (head, left/right shoulders, left/right elbows, left/right wrists, left/right hips). Frame-level ground-truth labels (stable/unstable) are annotated by Kinesiology experts from Cal Poly Pomona.

III-B. System Overview

As shown in Fig. 1, the system maps each 1-second rider-pose segment $\mathbf{x}$ to a binary label (stable/unstable). Two branches process each window in parallel: HCGAE encodes frames through three anatomical part graphs, while an adversarially trained GCN backbone provides classifier features. A lightweight MLP fusion head combines the two outputs and produces the final logit. Training proceeds in 2 stages: (i) pretrain / cache HCGAE and the GCN backbone. (ii) train only the fusion head with both frozen branches from i.

III-C. Body-Informed GCN Design

GCN backbone. The GCN backbone f_{θ}^{GCN} stacks two graph convolutional layers with improved self-loop variant [6]. It operates on a spatio-temporal graph $\mathcal{G}_{st}$, which consists of 270 node features (30 frames $\times$ 9 joints) and connects anatomical joints within each frame and links each frame to its consecutive frames. These node features are global-mean-pooled and passed through a three-layer MLP head. This architecture is used as a baseline model and also as the frozen backbone within HCGCN. The weights are θ^* , obtained by prior adversarial training under the protocol of Section III-D, and the backbone contributes the logit pair $\ell = f_{\theta^*}^{\text{GCN}}(\mathbf{x}) \in \mathbb{R}^2$ to the fusion head.

HCGAE. The HCGAE (Hard-Constraint Graph Autoencoder) encodes perturbation-stable anatomical features by compressing each pose frame into a latent code $\mathbf{h} \in \mathbb{R}^d$ ($d=16$).

These components are designed to characterize a plausible body structure, that is, a near-horizontal hip alignment, a bounded shoulder-torso ratio, and a symmetric arm chain. Each anatomical encoder consists of two graph convolutions and a graph mean pooling, which produces a $c=32$ -dimensional embedding for each pose frame. A bottleneck MLP, $E_{\phi_{\text{enc}}}$, maps this vector to the latent space $\mathbf{h} \in \mathbb{R}^d$, and a symmetric MLP decoder, $D_{\phi_{\text{dec}}}$, reconstructs the 9×2 frame $\hat{\mathbf{x}}$. The autoencoder is trained on frames by minimizing $\mathcal{L}_{\text{rec}} = \frac{1}{2J} \|\hat{\mathbf{x}} - \mathbf{x}\|_2^2$.

HCGCN. HCGCN, the body-informed GCN, fuses the two branches described above: HCGAE captures anatomical structure that remains stable under perturbation, while the GCN backbone provides discriminative classification features trained on the spatio-temporal graph. Formally, for a segment $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$ with $T = 30$, the frozen HCGAE is applied per frame to obtain a latent sequence $\mathbf{h}_t \in \mathbb{R}^d$. Each segment is summarized along the temporal dimension as mean and maximum vectors, denoted as $\mathbf{h}_{\text{mean}}$ and $\mathbf{h}_{\text{max}}$. The same procedure is applied to the pre-trained latent codes statistics from the training set, obtaining vectors $\mathbf{z}_{\text{mean}}$ and $\mathbf{z}_{\text{max}}$. Together with the backbone logits, ℓ , these summaries form the fusion vector

$$\varphi(\mathbf{x}) = [\mathbf{h}_{\text{mean}}; \mathbf{h}_{\text{max}}; \mathbf{z}_{\text{mean}}; \mathbf{z}_{\text{max}}; \ell] \in \mathbb{R}^{4d+2} = \mathbb{R}^{66}. \quad (1)$$

The trainable component of the HCGCN is a small fusion head $g_{\psi} : \mathbb{R}^{66} \rightarrow \mathbb{R}^2$, a three-layer MLP. Only ψ is updated during training; the parameters for HCGAE and the GCN backbone are frozen. A cross-entropy loss is used to train HCGCN:

$$\mathcal{L}_{\text{CE}}(\mathbf{x}) = - \sum_{c=1}^2 y_c \log(g_{\psi}(\varphi(\mathbf{x}))_c) \quad (2)$$

III-D. Model Training

Adversarial training. In each training batch, we build a worst-case perturbed copy of the input against the model's current

Algorithm 1 One outer epoch of dynamic adversarial training on HCGCN.

Require: trainable head g_ψ ; frozen $E_\phi, f_{\theta^*}^{\text{GCN}}$; batches $\{(\mathbf{X}, \mathbf{Y})\}$; mixing r ; PGD parameters (ε, K, α) ; optimizer ADAMW.

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1: for each batch  $(\mathbf{X}, \mathbf{Y})$  do
2:   model.fusion_head.eval()
3:    $\tilde{\mathbf{X}} \leftarrow \text{PGD-K}(f^{\text{HCGCN}}, \mathbf{X}, \mathbf{Y}, \varepsilon, K, \alpha)$   $\triangleright$  Eq. (3)
4:   model.fusion_head.train()
5:    $\mathcal{L} \leftarrow (1-r)\mathcal{L}_{\text{CE}}(\mathbf{X}) + r\mathcal{L}_{\text{CE}}(\tilde{\mathbf{X}})$   $\triangleright$  Eq. (2)
6:    $\psi \leftarrow \text{ADAMW.STEP}(\nabla_\psi \mathcal{L})$ 
7: end for
8: COSINEANNEALINGLR.STEP()

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weights, then train on a 50/50 mix of clean and perturbed examples. The perturbation itself is an untargeted K -step projected-gradient attack in the L^∞ norm. Starting from a random $\mathbf{x}^{(0)}$ inside the ε -ball around $\mathbf{x}$, the inner loop of the PGD iterates in this way

$$\mathbf{x}^{(i+1)} = \Pi\left(\mathbf{x}^{(i)} + \alpha \cdot \text{sgn}\left(\nabla_{\mathbf{x}^{(i)}} \mathcal{L}_{\text{CE}}(\mathbf{x}^{(i)})\right)\right). \quad (3)$$

where Π projects elementwise into the valid ε -bounded input range after each PGD step. [5]. The step size is $\alpha = \varepsilon/K$ and the perturbation budget is $\varepsilon = 0.03$. Let η denote the trainable parameter block (the full θ for the plain GCN and ψ for HCGCN), the training objective [8] is

$$\min_{\eta} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_{\text{train}}} [(1-r)\mathcal{L}_{\text{CE}}(\mathbf{x}) + r\mathcal{L}_{\text{CE}}(\tilde{\mathbf{x}})], \quad (4)$$

with mixing ratio $r = 0.5$ and $\tilde{\mathbf{x}}$ generated by Equation (3) against the current η per batch. Algorithm 1 states one outer epoch of the adversarial training procedure for HCGCN.

Hyperparameters. We set training- $\varepsilon = 0.03$ based on a preliminary grid search over ε , PGD steps K , and mixing ratio r . We fix $r = 0.5$ for both plain GCN and HCGCN. The four attacks are the same method run for an increasing number of inner steps: RFGSM is the one-step random-start variant ($K=1$), and PGD-7, PGD-10, PGD-15 use $K=7, 10, 15$. During adversarial training, K is chosen to match the attack strategy with a perturbation budget of $\varepsilon = 0.03$. At the evaluation time, all models are tested across the full range of the attack-step ladder using $\varepsilon = 0.01$.

All models are trained with ADAMW, learning rate 3×10^{-3} , weight decay 10^{-3} , cosine annealing, and batch size 32. Plain GCN is trained for 50 epochs, while HCGCN is trained for fewer epochs (25) as faster convergence is observed.

IV. Evaluation

IV-A. Experimental Setup

For each of the two architectures, HCGCN and the plain GCN, we train one clean baseline and four adversarially trained variants corresponding to RFGSM, PGD-7, PGD-10, and PGD-15. All results are reported as mean $\pm$ standard deviation over five independent runs. The main metric is test ROC-AUC at $\varepsilon = 0.01$; adversarial training uses $\varepsilon = 0.03$.

TABLE I

ROUND 9 LEAVE-ONE-ATTACK-OUT TEST ROC-AUC AT EVALUATION $\varepsilon = 0.01$ (TRAINING $\varepsilon = 0.03$), 5-SEED MEAN $\pm$ STD. HELD-OUT ROWS: THE ARCHITECTURE IS TRAINED ON A UNIFORM MIX OF THE OTHER THREE ATTACKS AND EVALUATED ON THE NAMED ATTACK. “CLEAN” AVERAGES CLEAN-TEST PERFORMANCE ACROSS FOUR LOAO MODELS.

	GCN	HCGCN	Δ
Clean	0.604	0.918	+0.314
RFGSM	0.412 $\pm$.048	0.807 $\pm$.027	+0.395
PGD-7	0.285 $\pm$.079	0.704 $\pm$.035	+0.419
PGD-10	0.447 $\pm$.053	0.670 $\pm$.014	+0.223
PGD-15	0.328 $\pm$.075	0.645 $\pm$.013	+0.317
Mean held-out	0.368	0.707	+0.339

IV-B. Single-Attack Robustness

We summarize the results as a 5×5 attack-transfer matrix for each model variant of the baseline plain GCN and HCGCN trained with different attacks in Fig. 2.

Clean-side cost. HCGCN preserves clean performance under adversarial training, while plain GCN shows a severe clean-accuracy trade-off. In the first column of Fig. 2, adversarial training drops the clean ROC-AUC of plain GCN from 0.97 to about 0.56, indicating a substantial loss of clean-data discriminative power. In contrast, HCGCN remains stable with a clean ROC-AUC of 0.96 before adversarial training and about 0.92 after adversarial training. This indicates that HCGCN gains adversarial robustness without losing clean-data performance.

Per-attack robustness. HCGCN also shows stronger adversarial robustness across attacks. Among the adversarially trained models, HCGCN-AT-PGD-15 achieves an average adversarial ROC-AUC of 0.74 across the four evaluation attacks, compared with 0.35 for GCN-AT-PGD-15. Further, HCGCN-AT outperforms GCN-AT in every attack-evaluation cell at the five-seed mean. The off-diagonal AT cells further show that HCGCN trained from one attack retains robustness against other attacks, suggesting better cross-attack generalization.

Seed-level reliability. HCGCN-AT seed-level standard deviations stay ≤ 0.06 on every adversarial cell of Figure 2, against up to 0.14 for GCN-AT: the body-informed defense is also the more reproducible of the two.

IV-C. Cross-Attack Generalization

We also test the harder regime: train on three of the four attacks and test on the one held out. For each held-out attack $a \in \{\text{RFGSM}, \text{PGD-7}, \text{PGD-10}, \text{PGD-15}\}$, one adversarial model is trained against a uniform mix of the other three attacks and evaluated on all four. Each batch samples one of the three training attacks and runs it at its native step count. The protocol otherwise matches Section IV-A: five seeds, evaluation $\varepsilon = 0.01$ (training $\varepsilon = 0.03$). Table I reports the per-family LOAO results.

Held-out generalization. HCGCN-LOAO holds a mean held-out adversarial ROC-AUC of 0.707 across the four held-out attacks, with a cell-level range of 0.645 ± 0.013 on PGD-15 to 0.807 ± 0.027 on RFGSM. The plain-GCN family reaches 0.368 under the same protocol. HCGCN-LOAO beats

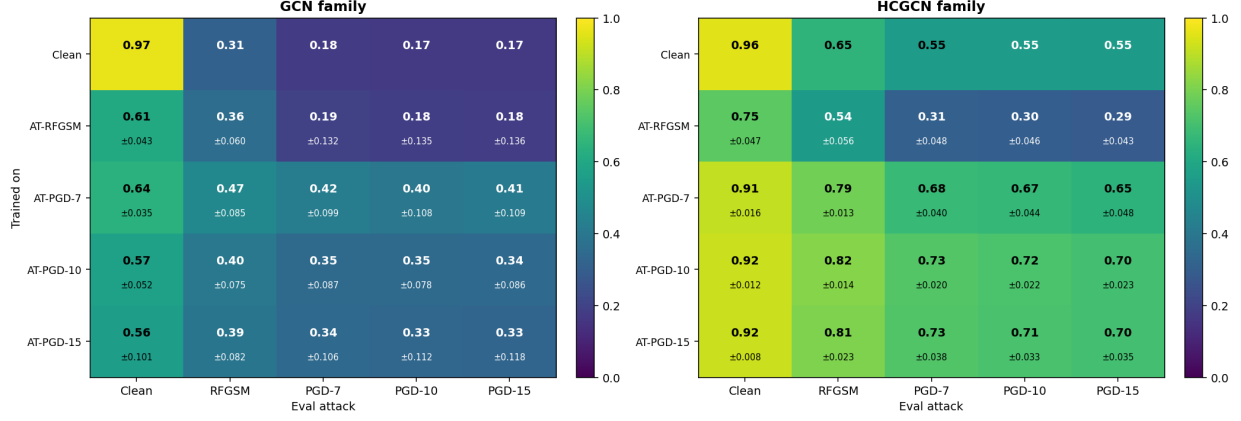
Single-attack AT generalization (AUC at $\epsilon=0.01$, 5-seed mean $\pm$ std)

Fig. 2. Attack-vs-attack generalization at evaluation $\epsilon = 0.01$ (training $\epsilon = 0.03$), 5-seed mean $\pm$ std test ROC-AUC. Rows are what each model was trained on (Clean baseline or an adversarially-trained AT- X variant); columns are what it was tested against (Clean inputs or one of the four attacks). Diagonal AT-row cells are in-distribution (same attack for train and eval); off-diagonal AT-row cells are cross-attack. Left panel: plain-GCN family. Right panel: HCGCN family.

TABLE II

HCGAE PART-ENCODER ABLATION UNDER THE SECTION IV-C LOAO PROTOCOL. TRAINING AND EVALUATION BOTH AT $\epsilon = 0.03$, SEED 67. TOP 4 VARIANTS BY HELD-OUT MEAN ROC-AUC. “MEAN” IS THE HELD-OUT ADVERSARIAL ROC-AUC AVERAGED OVER FOUR HELD-OUT ATTACKS.

	k	Clean	Mean adv	Δ
HCGAE (ours)	3	0.957	0.497	—
only_hip_torso	2	0.959	0.480	−0.017
no_shoulder_torso	3	0.958	0.473	−0.024
no_hip_line	3	0.959	0.463	−0.034
only_torso_arms	2	0.958	0.461	−0.036

GCN-LOAO on every held-out cell of Table I at 5-seed mean. On PGD-7, PGD-10, and PGD-15, the HCGCN-LOAO held-out cells sit within 0.06 ROC-AUC of their single-attack counterparts in Fig. 2 (worst-case drop 0.058 on PGD-15, $0.703 \rightarrow 0.645$); the held-out RFGSM cell at 0.807 substantially exceeds the single-attack RFGSM cell at 0.545. HCGCN-LOAO retains a clean ROC-AUC of 0.918, against 0.957 for the clean HCGCN baseline.

Seed-level reliability. Variance follows the same ordering as in Section IV-B: across all held-out sets in Table I, HCGCN-LOAO stays at ≤ 0.06 , versus up to 0.08 for GCN-LOAO.

IV-D. Ablation Study

At seed 67, we compare HCGCN’s choice of three anatomical part-encoders against alternative subsets of HCGAE’s four candidate encoders under the LOAO protocol of Section IV-C, denoting each variant by the number of active encoders k . Table II lists the top 4 variants by held-out mean ROC-AUC; HCGCN’s 3-part configuration is the best across the lattice.

V. Conclusion

We presented HCGCN, a body-informed defense for pose-based rider-stability classification that fuses a frozen anatomically disentangled HCGAE with an adversarially-trained GCN

backbone through a small trainable fusion head. Extensive experiment results on real-world mobility scooter riding data have shown that HCGCN improves robustness compared to both the plain GCN and the adversarially trained GCN baseline models, and maintains a high clean accuracy.

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